The Value of Visualization + Interaction

Alex Kale

Slides credited to

Jessica Hullman

Estimated 175 zettabytes in 2025
(175,000,000,000,000,000,000,000 bytes)

Estimated 612 by 2030 because of AI

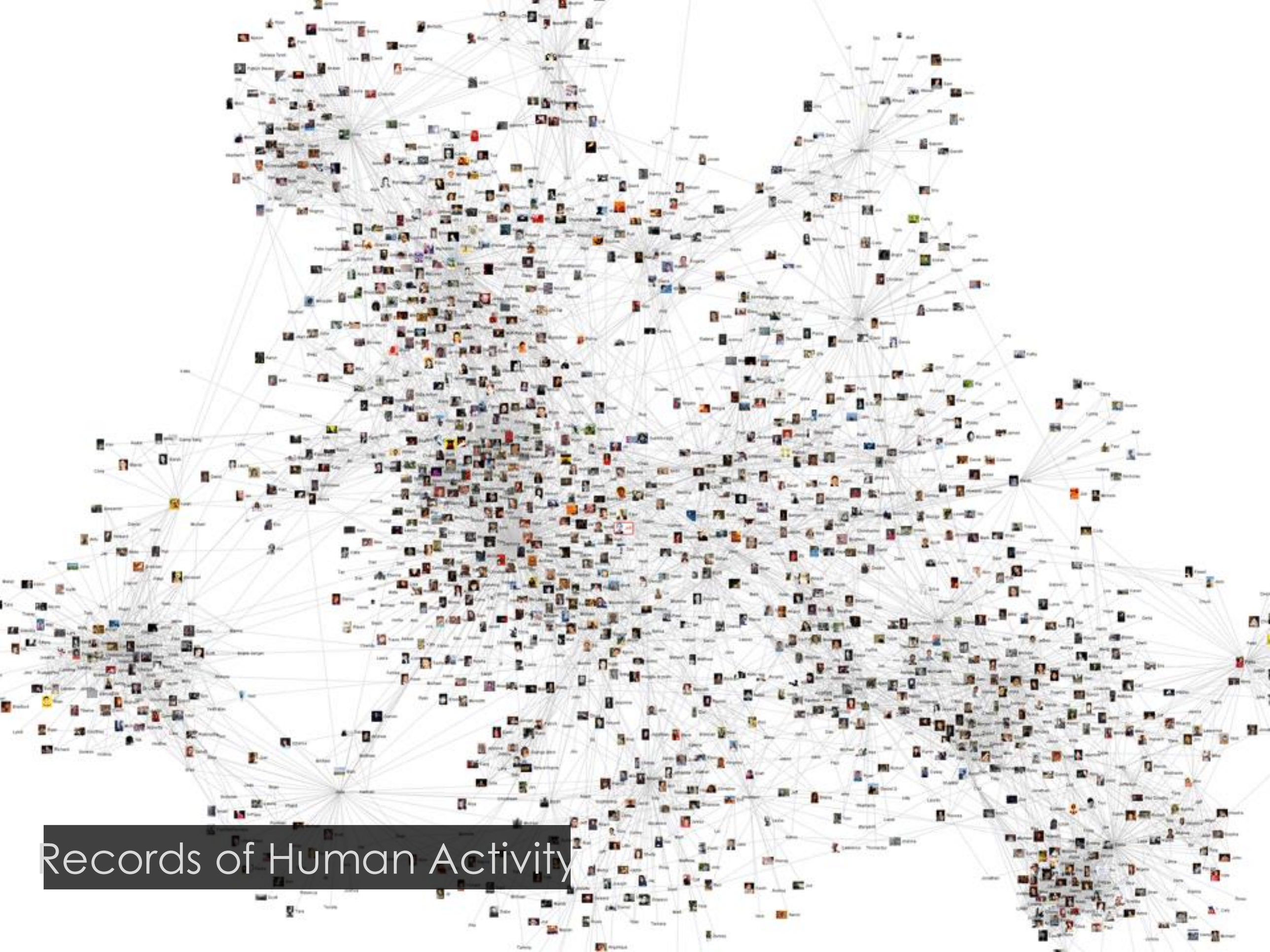
"If you attempted to download 175 zettabytes at the average current internet connection speed, it would take you 1.8 billion years to download. Even if you enlisted every person in the world to help with the download, it would still take 81 days."

https://www.bernardmarr.com/default.asp?contentID=1846

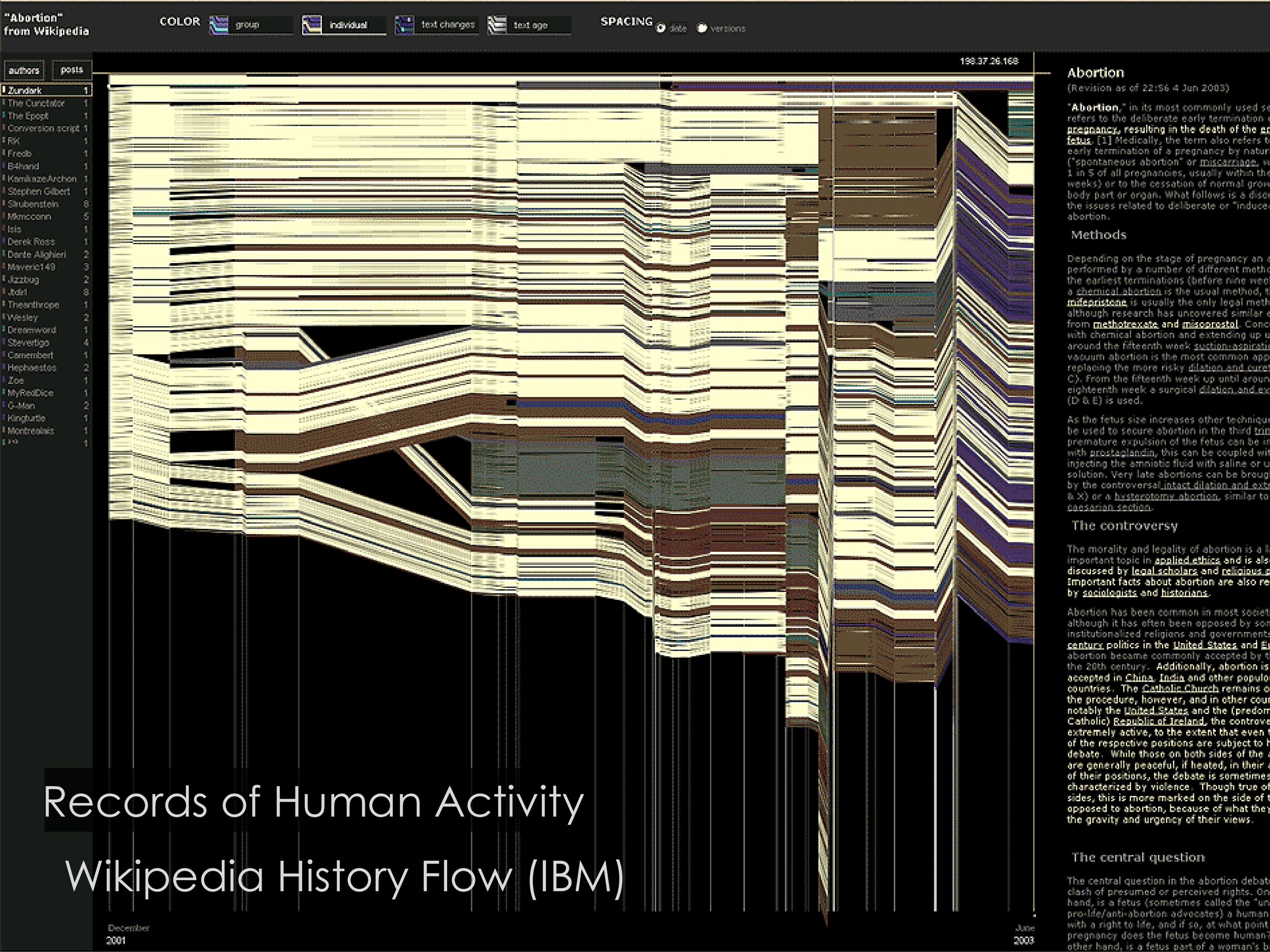
Physical Sensors

Image: toddwschneider.com

Health & Medicine



Records of Human Activity



Records of Human Activity

Wikipedia History Flow (IBM)

“What information consumes is rather obvious: **it consumes the attention of its recipients**. Hence a wealth of information creates a poverty of attention, and a need to allocate that attention efficiently among the overabundance of information sources that might consume it.”

~Herb Simon
as quoted by Hal Varian
Scientific American
September 1995

What is visualization?

What is visualization?

"Transformation of the symbolic into the geometric"
[McCormick et al. 1987]

"... finding the artificial memory that best supports our natural means of perception." [Bertin 1967]

"The use of computer-generated, interactive, visual representations of data to amplify cognition."
[Card, Mackinlay, & Shneiderman 1999]

Information Visualization

“The use of computer-generated, interactive, visual representations of data abstract to amplify cognition.”

(In contrast, visualization where the mapping is given is more often called scientific visualization)

Set A

X	Y
10	8.04
8	6.95
13	7.58
9	8.81
11	8.33
14	9.96
6	7.24
4	4.26
12	10.84
7	4.82
5	5.68

Set B

X	Y
10	9.14
8	8.14
13	8.74
9	8.77
11	9.26
14	8.1
6	6.13
4	3.1
12	9.11
7	7.26
5	4.74

Set C

X	Y
10	7.46
8	6.77
13	12.74
9	7.11
11	7.81
14	8.84
6	6.08
4	5.39
12	8.15
7	6.42
5	5.73

Set D

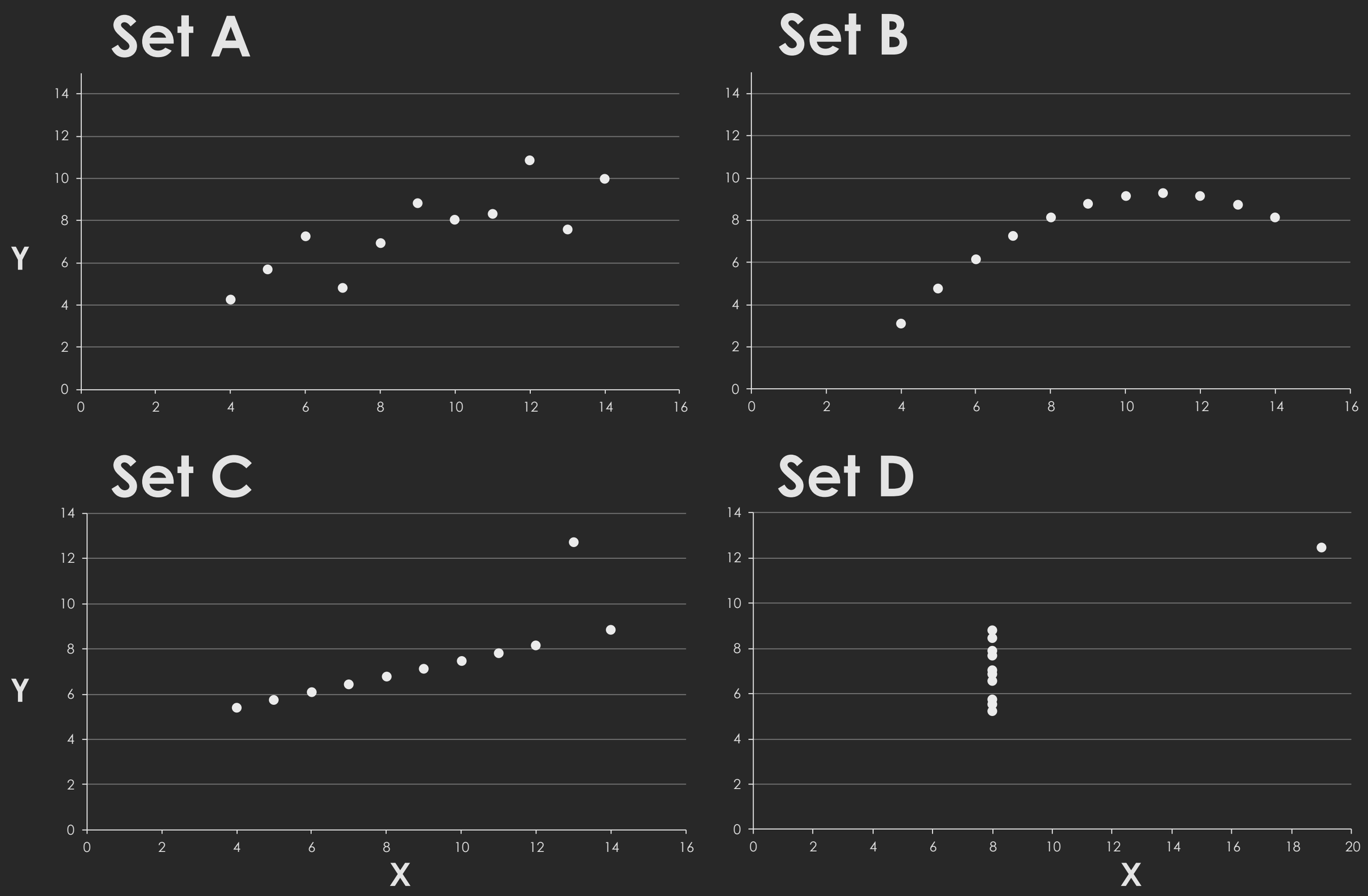
X	Y
8	6.58
8	5.76
8	7.71
8	8.84
8	8.47
8	7.04
8	5.25
19	12.5
8	5.56
8	7.91
8	6.89

Summary Statistics

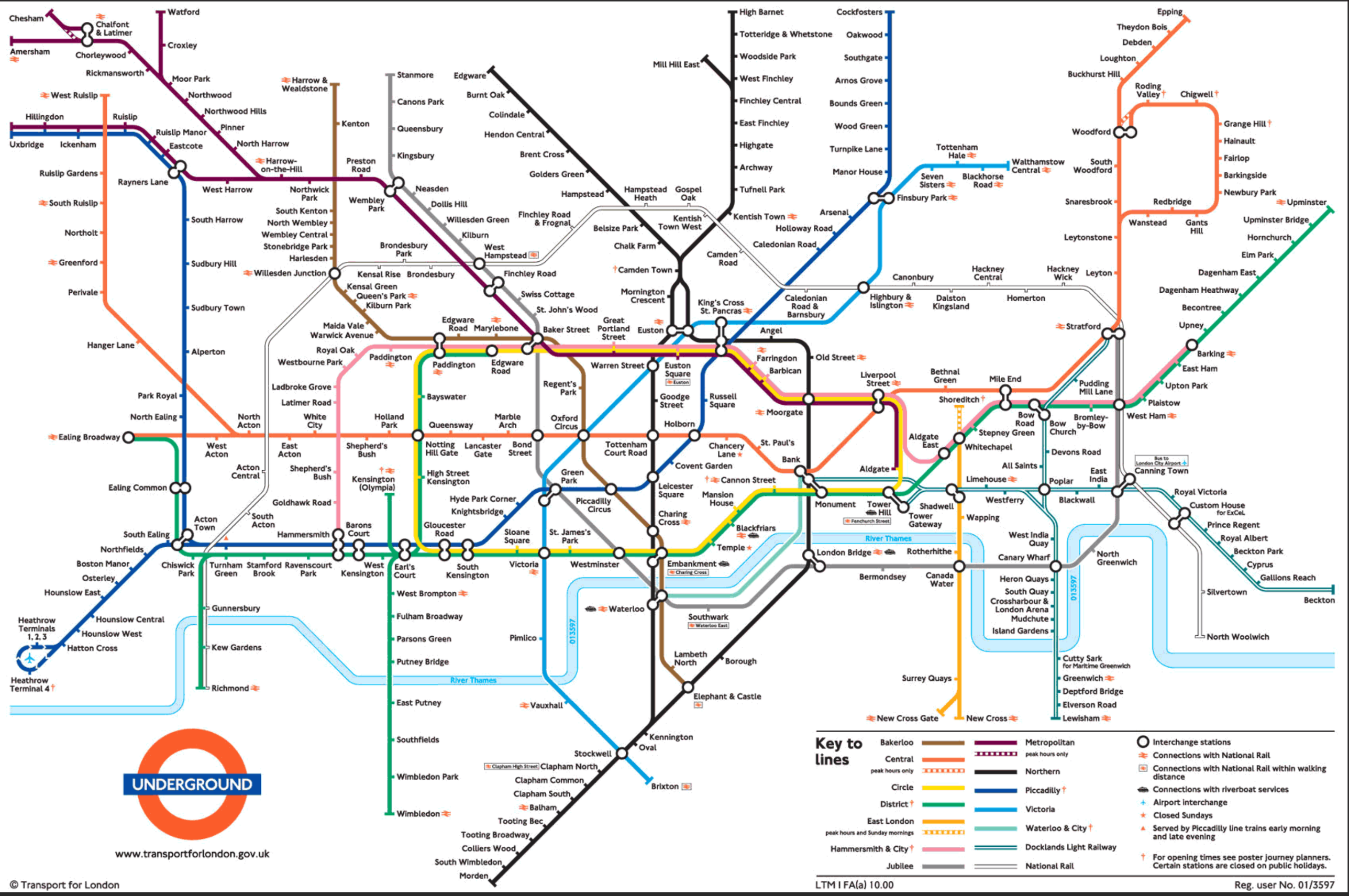
$u_X = 9.0$ $\sigma_X = 3.317$ $Y = 3 + 0.5\ X$

$u_Y = 7.5$ $\sigma_Y = 2.03$ $R^2 = 0.67$

[Anscombe 73]

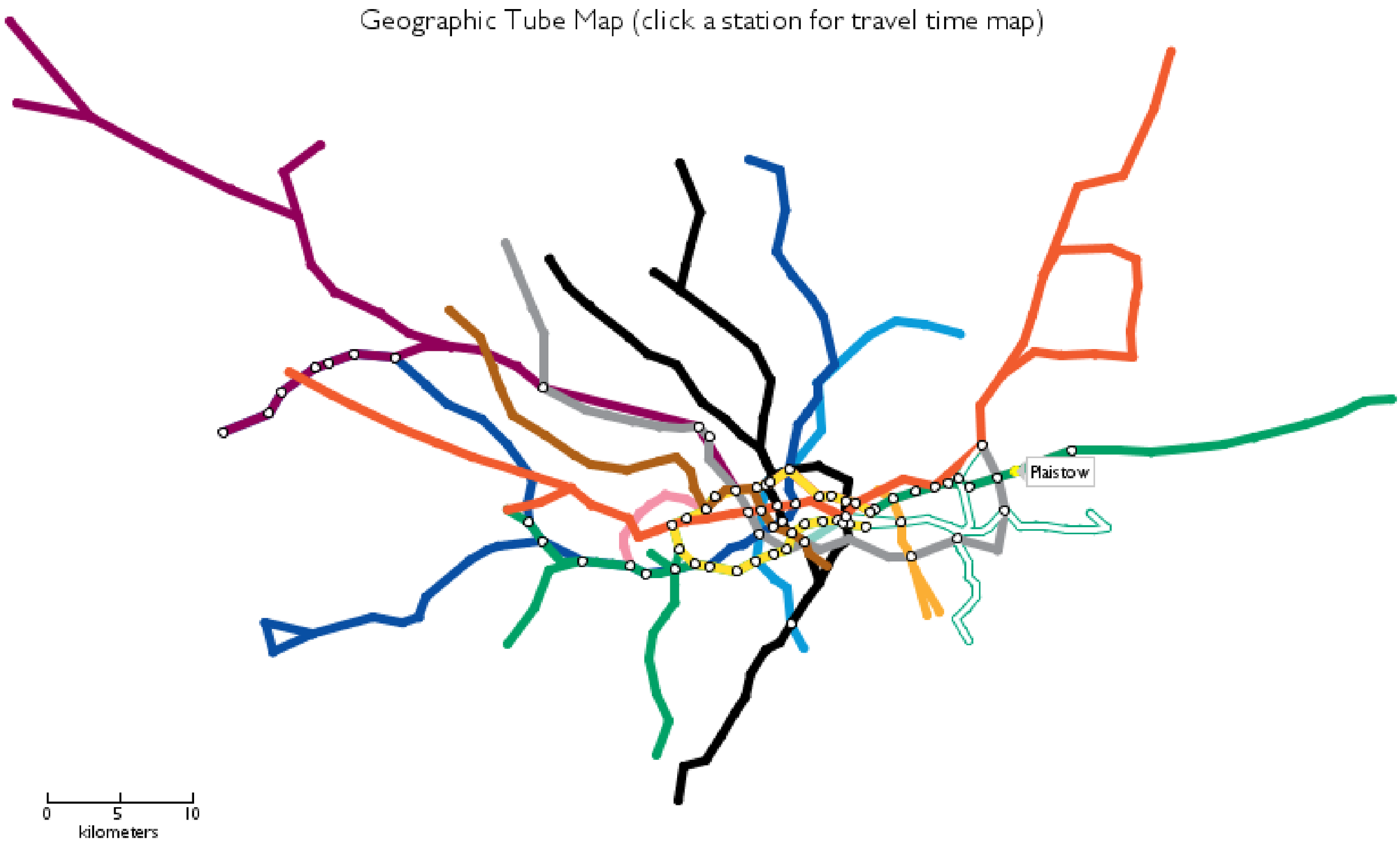

Set A
Set B
Set C
Set D
Y
X
0
2
4
6
8
10
12
14
16
18
20

Why interactive/dynamic?



© Transport for London

London Tube



By Tom Carden, for more info see
http://www.creativeapplications.net/processing/travel-time-tube-map-processing/

tube_map_travel_times_js
Geographic Tube Map (click a station for travel time map)
0
5
10
kilometers

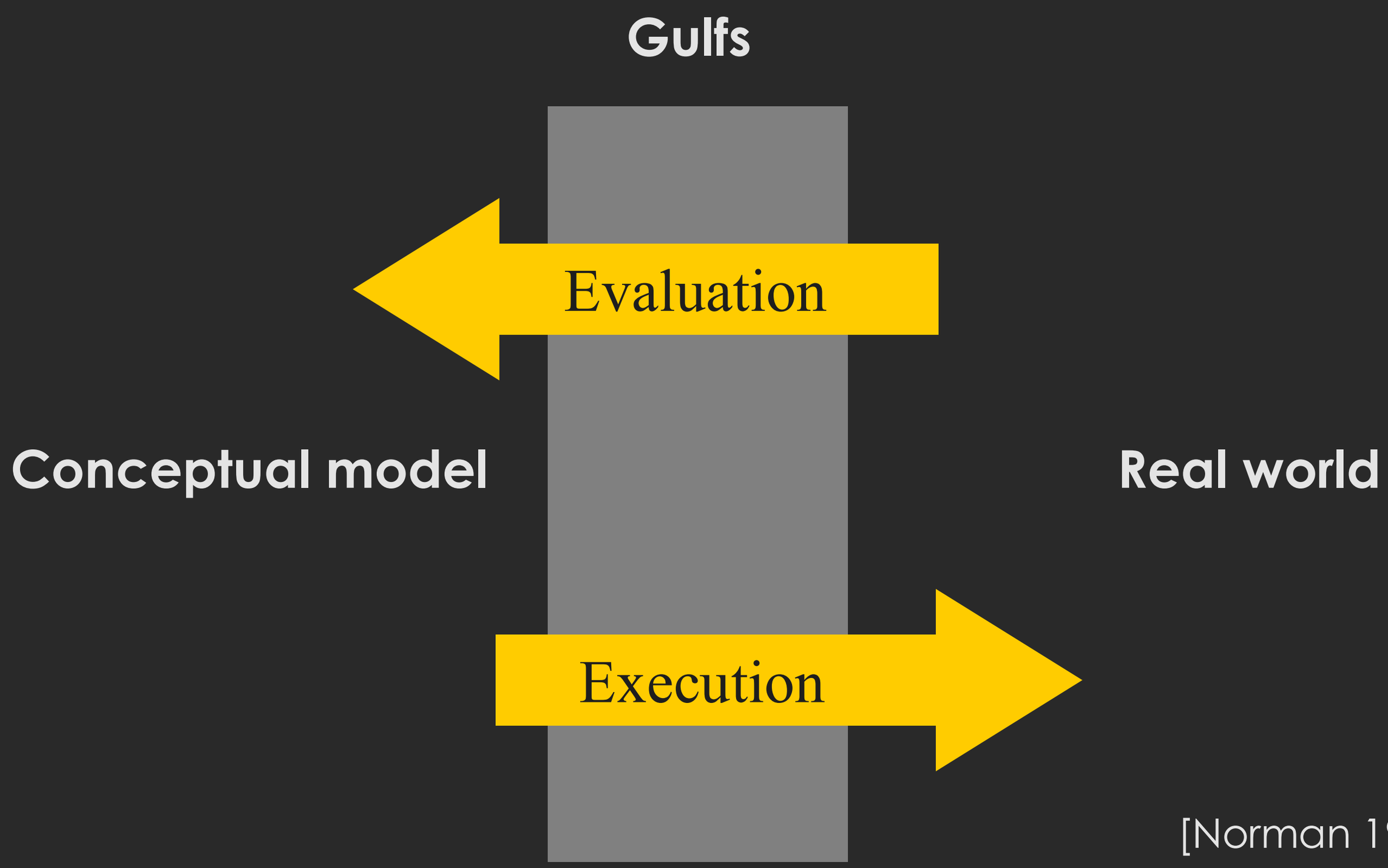
Interactivity is a multiplier

increases number of supported tasks

increases number of comparisons

makes comparisons easier

Gulfs of execution & evaluation



[Norman 1986]

Gulfs of execution & evaluation

Gulf of Execution

The difference between the user's intentions and the allowable actions.

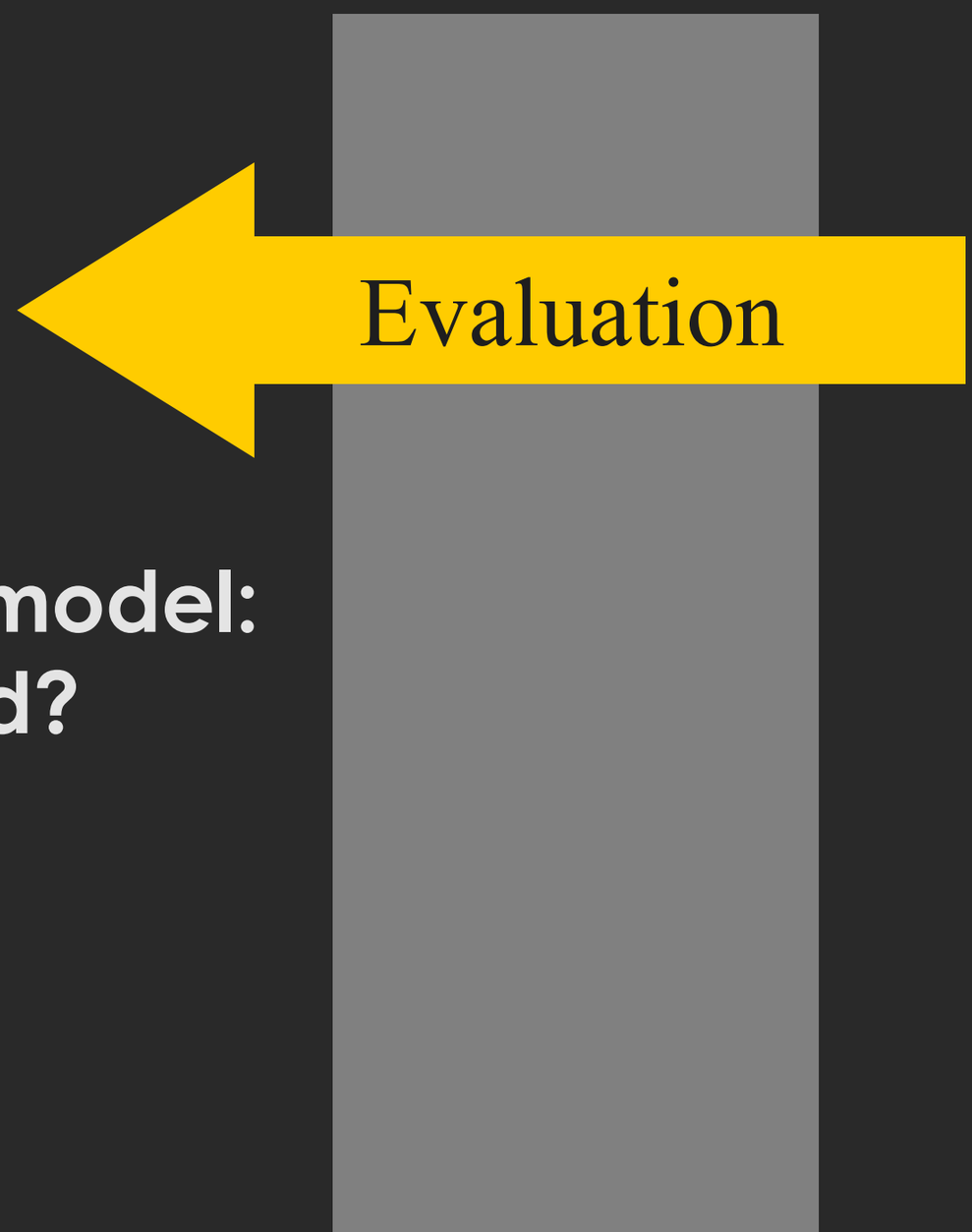
Gulf of Evaluation

The amount of effort that the person must exert to interpret the state of the system and to determine how well the expectations and intentions have been met.

[Norman 1986]

Gulf of evaluation

Gulf



Conceptual model:
x,y correlated?

Real world:

X	Y
0.67	0.79
0.32	0.63
0.39	0.72
0.27	0.85
0.71	0.43
0.63	0.09
0.03	0.03
0.20	0.54
0.51	0.38
0.11	0.33
0.46	0.46

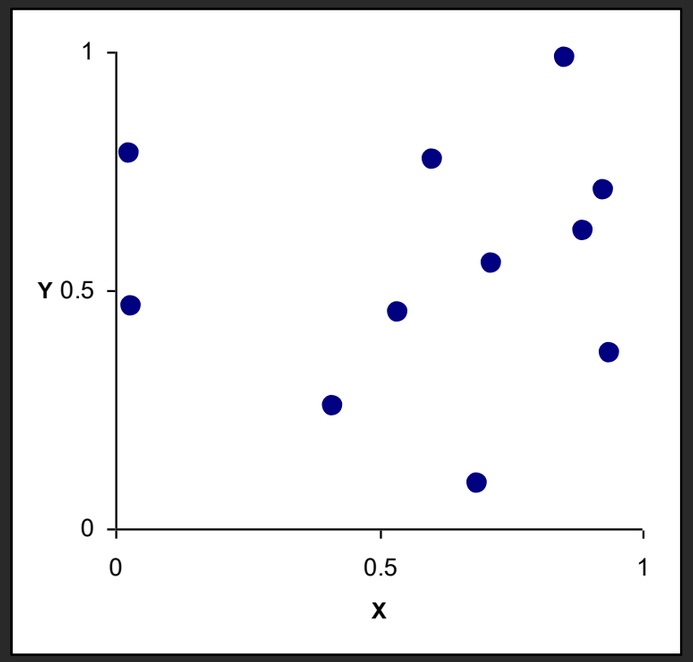
Gulf of evaluation

Gulf

Evaluation

Conceptual model:
x,y correlated?

Real world:



Gulf of evaluation

Gulf

Evaluation

Real world:

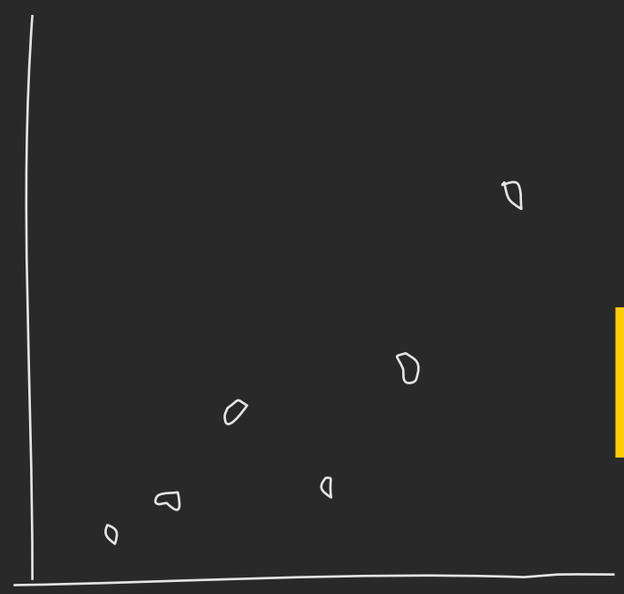
$\rho = -.29$

Conceptual model:
x,y correlated?

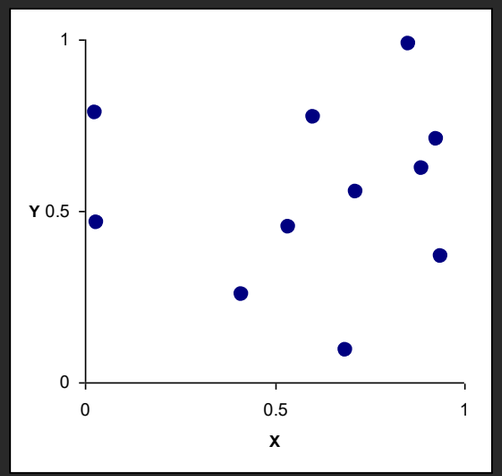
Gulf of execution

Gulf

Conceptual model:
Draw a scatterplot



Execution



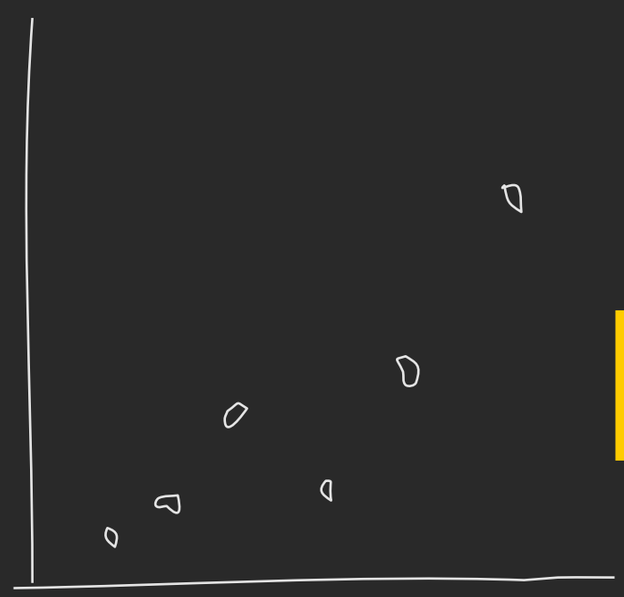
Real world

Move 90 30
Rotate 35
Pen down
…

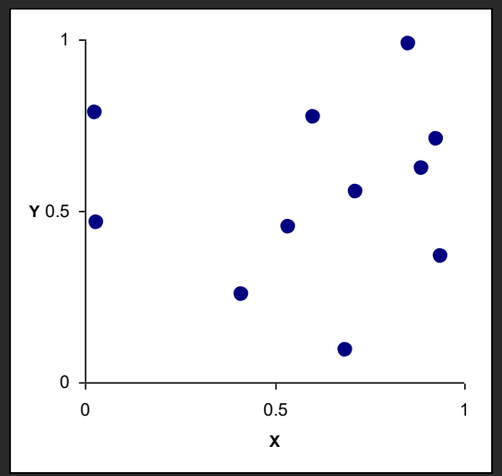
Gulf of execution

Gulf

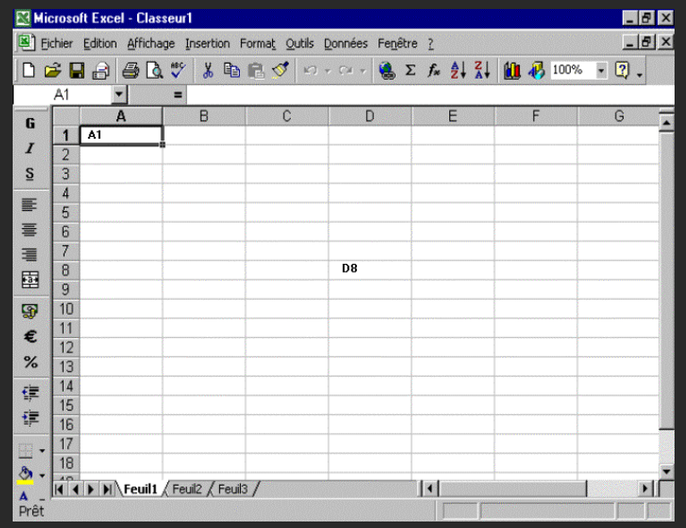
Conceptual model:
Draw a scatterplot



Execution



Real world



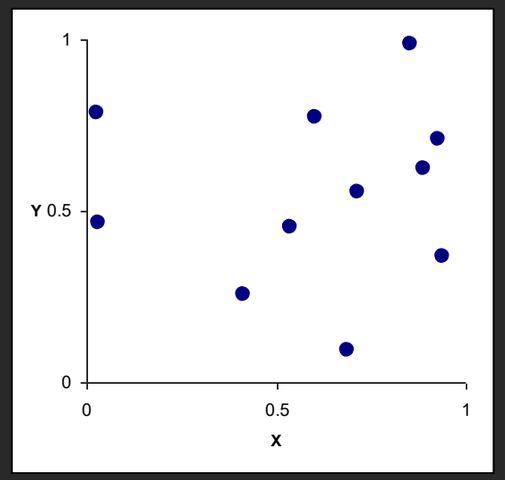
Gulf of envisioning

Gulf

Conceptual model:
Draw a scatterplot

Prompt

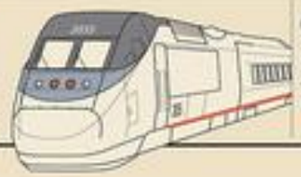
Envisioning



Infographics
(a brief note)

LA FACCIA DELLA RAPIDITÀ

Alcuni esempi di treni ad alta velocità attualmente operativi. Per ciascuno è indicato il Paese, il nome, l'anno di entrata in servizio e la velocità massima



Acela
STATI UNITI
Acela Express
2000
250 km/h

ITALIA
Frecciarossa - Etr 500 Av
2008
300 km/h

SNCF
FRANCIA
Tgv Duplex
1995
320 km/h

CRH
CINA
Crh2
2008
350 km/h

N700
GIAPPONE
N700 (Shinkansen)
2007
300 km/h

Più veloce dell'aereo

Il Giappone progettava le prime linee rapide già negli anni Cinquanta. Oggi i binari jet nel mondo sono quasi 10 mila km. Ma nel 2025 saranno il triplo e ruberanno i passeggeri al cielo

– di **Francesco Franchi**

LINEE VELOCI NEL MONDO

La mappa mostra i collegamenti ferroviari mondiali ad alta velocità (≥250 km/h). Le corone circolari indicano, per ogni Stato, **la lunghezza totale** e **la velocità media** delle tratte in questione. Più l'arco è spesso, maggiore è la velocità; più l'arco è ampio, maggiore è la lunghezza.

- Chilometri operativi
- Chilometri in costruzione
- Chilometri in progetto
- Tratte operative
- Tratte in costruzione e in progetto
- Aggiornamento linee tradizionali

CHILOMETRI TOTALI AD ALTA VELOCITÀ

ESISTENTI · IN COSTRUZIONE · IN PROGETTO · PREVISIONE 2025

	Esistenti	In costruzione	In progetto	Previsione 2025
Europa	5.508	3.474	8.501	17.523
Asia	3.050	4.825	7.857	16.657
Altri Stati	302	2.335	2.251	
Mondo	9.015	8.295	18.753	36.987

FONTE: Uic (Union international chemin de fer)

1956 Il Giappone progetta una linea veloce (250 km/h) tra Tokyo e Osaka. Viene inaugurata nel 1964

1970 Iniziano i lavori per la Direttissima tra Roma e Firenze. L'intera tratta è ultimata nel 1992

1976 La Sncf francese lancia il progetto di linea veloce Parigi-Lione. Entra in funzione nel 1981

1991 Primi InterCityExpress in Germania: Hannover-Würzburg e Mannheim-Stoccarda

1992 Diventa operativa la linea tra Madrid e Siviglia: prima tratta dell'alta velocità in Spagna

1999 In Giappone il treno Maglev (a levitazione magnetica) tocca in un test la velocità record di 552 km/h

2000 Negli Usa inizia a funzionare il collegamento veloce tra New York, Washington e Boston

2008 La Cina inaugura la linea Pechino-Tianjin: 120 km in 29 minuti. In Italia apre la Tav Milano-Bologna

38 *% è l'aumento degli investimenti cinesi per l'alta velocità nel primo semestre 2008*

39

http://www.flickr.com/photos/ffranchi/3199102299/sizes/l/in/pool-16135094@N00/

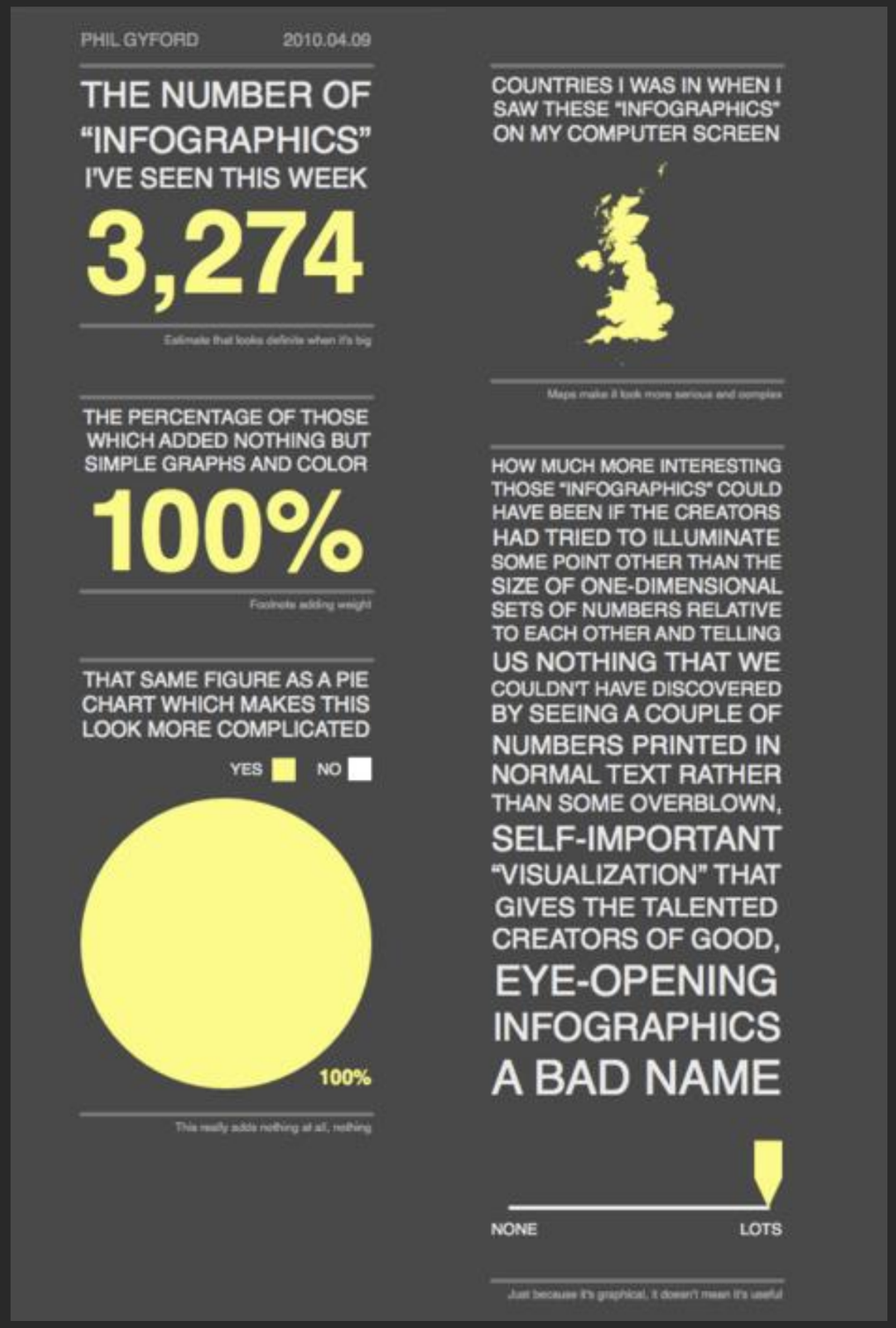
Infographics

Tend to be:

- Elaborate
- Descriptive
- Self-contained
- Frequently static

May use infovis techniques, but often the focus is instead graphic design

Not covered in this class



Why do we create visualizations?

Three functions of visualizations

Record information

- Photographs, blueprints, diagrams …

Support reasoning about information (analyze)

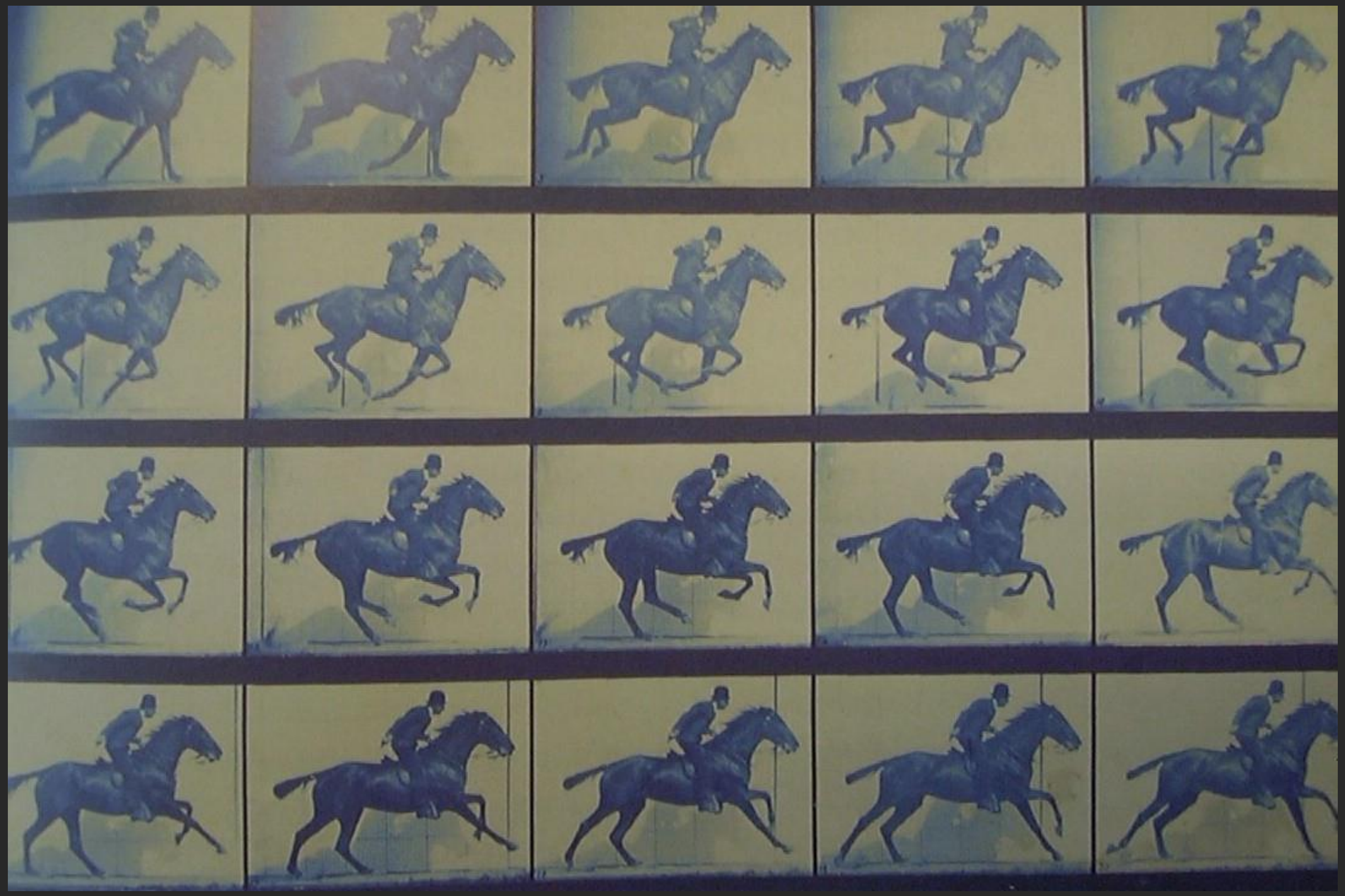
- Process and calculate
- Reason about data
- Feedback and interaction

Convey information to others (present)

- Share with other people
- Collaborate and revise
- Emphasize important aspects of data

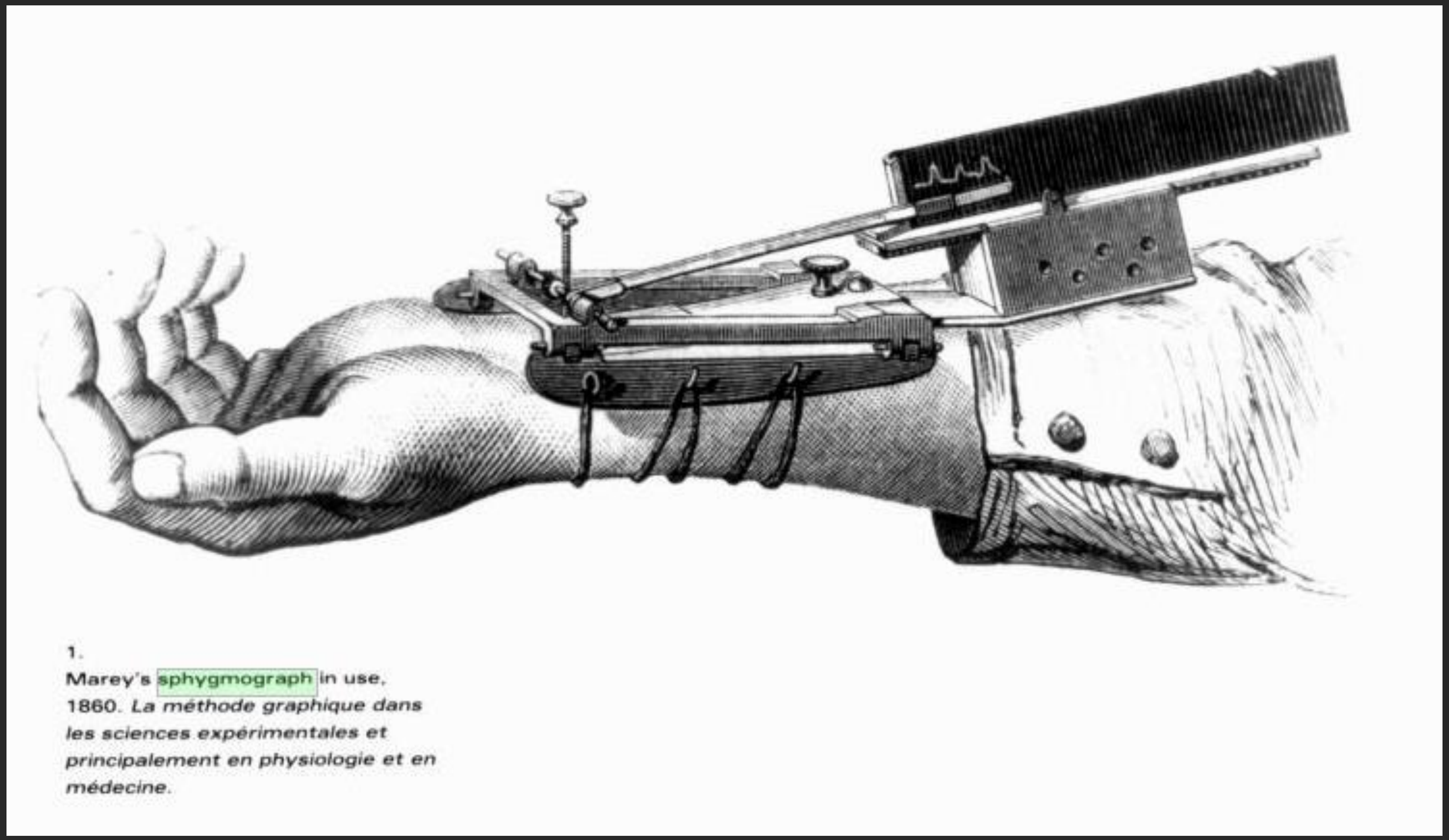
Record Information

Answer question



Gallop, Bay Horse “Daisy” [Muybridge 1884-86]

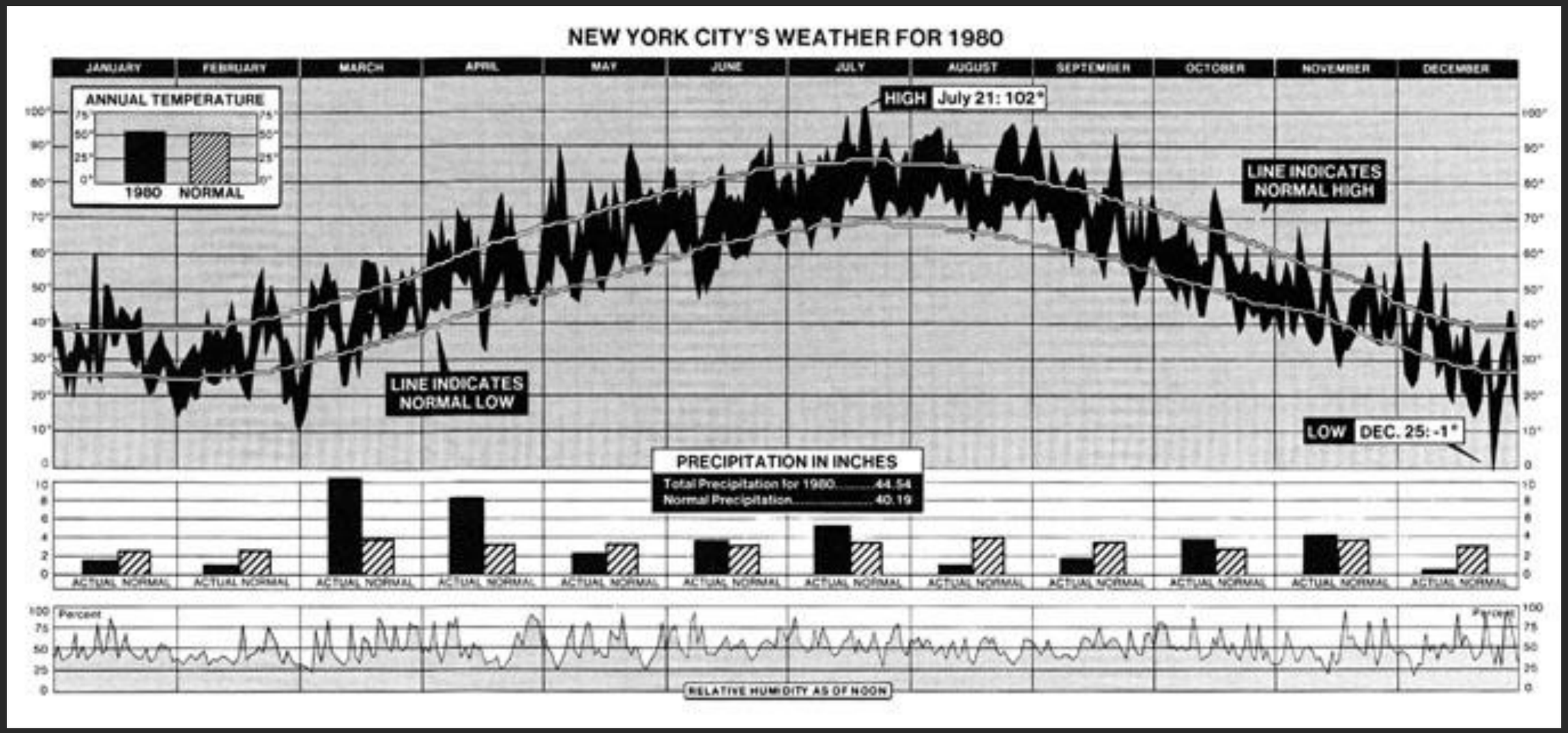
Other recording instruments



Marey's sphygmograph [from Braun 83]

Support Reasoning

Find patterns: New York weather

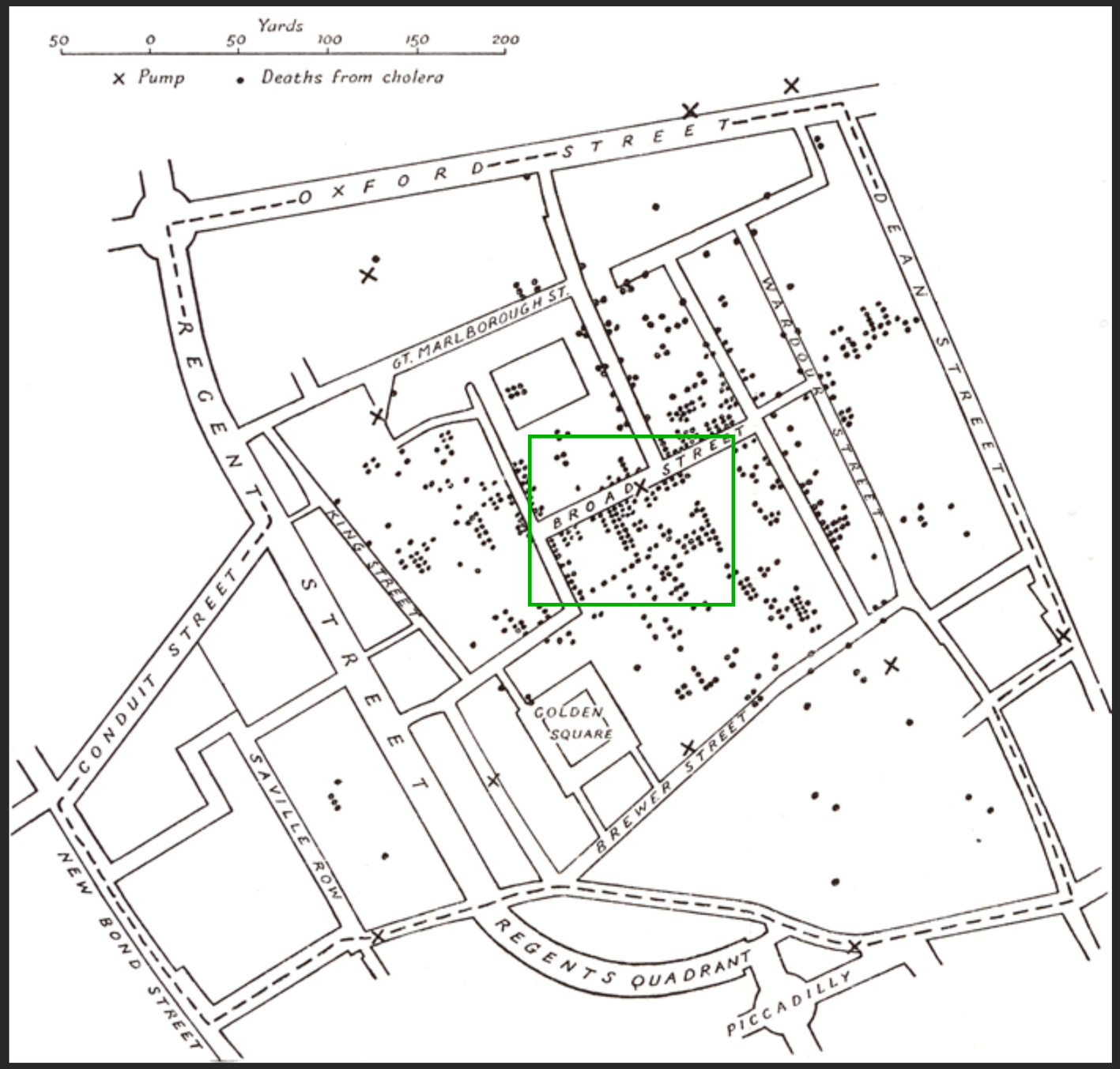


From the New York Times 1981

Answer a question: Cholera outbreak

In 1854 John Snow plotted the position of each cholera case on a map. [from Tufte 83]

Answer a question: Cholera outbreak

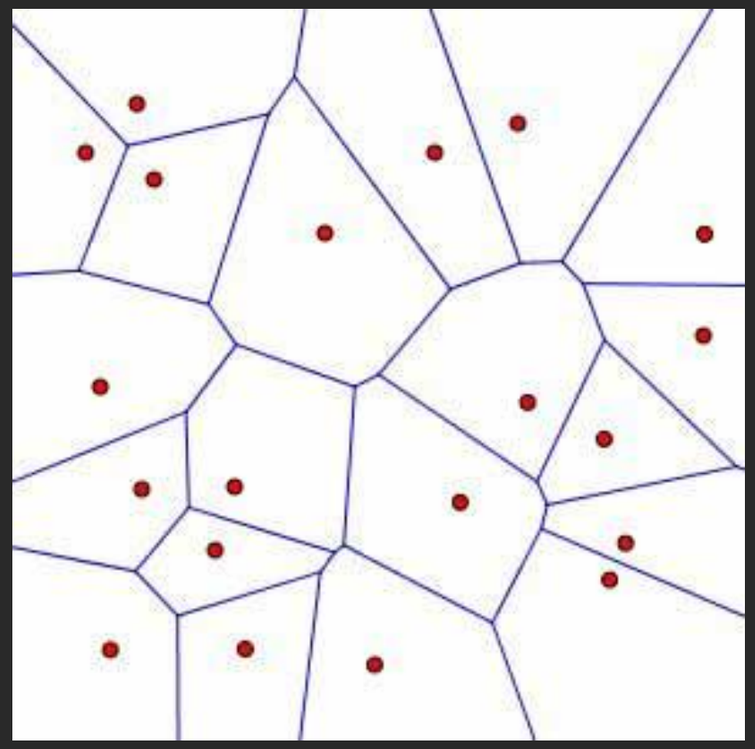


Used map to support hypothesis Broad St. pump was the cause. [from Tufte 83]

Answer a question: Cholera outbreak

The curve marks points at equal distance from the Broad Street pump and another pump.

The first Voronoi diagram



Convey Information to Others

Tell a story: Challenger

HISTORY OF O-RING DAMAGE ON SRM FIELD JOINTS

	SRM No.	Cross Sectional View: Erosion Depth (in.)	Cross Sectional View: Perimeter Affected (deg)	Cross Sectional View: Nominal Dia. (in.)	Top View: Length Of Max Erosion (in.)	Top View: Total Heat Affected Length (in.)	Clocking Location (deg)
61A LH Center Field**	22A	None	None	0.280	None	None	36°--66°
61A LH ~~CENTER~~ FIELD**	22A	NONE	NONE	0.280	NONE	NONE	338°-18°
51C LH Forward Field**	15A	0.010	154.0	0.280	4.25	5.25	163
51C RH Center Field (prim)***	15B	0.038	130.0	0.280	12.50	58.75	354
51C RH Center Field (sec)***	15B	None	45.0	0.280	None	29.50	354
41D RH Forward Field	13B	0.028	110.0	0.280	3.00	None	275
41C LH Aft Field*	11A	None	None	0.280	None	None	--
41B LH Forward Field	10A	0.040	217.0	0.280	3.00	14.50	351
STS-2 RH Aft Field	2B	0.053	116.0	0.280	--	--	90

*Hot gas path detected in putty. Indication of heat on O-ring, but no damage.
**Soot behind primary O-ring.
***Soot behind primary O-ring, heat affected secondary O-ring.

Clocking location of leak check port - 0 deg.

OTHER SRM-15 FIELD JOINTS HAD NO BLOWHOLES IN PUTTY AND NO SOOT NEAR OR BEYOND THE PRIMARY O-RING.

SRM-22 FORWARD FIELD JOINT HAD PUTTY PATH TO PRIMARY O-RING, BUT NO O-RING EROSION AND NO SOOT BLOWBY. OTHER SRM-22 FIELD JOINTS HAD NO BLOWHOLES IN PUTTY.

BLOW BY HISTORY

SRM-15 WORST BLOW-BY
- 2 CASE JOINTS (80°), (110°) ARC
- MUCH WORSE VISUALLY THAN SRM-22

SRM 22 BLOW-BY
- 2 CASE JOINTS (30-40°)

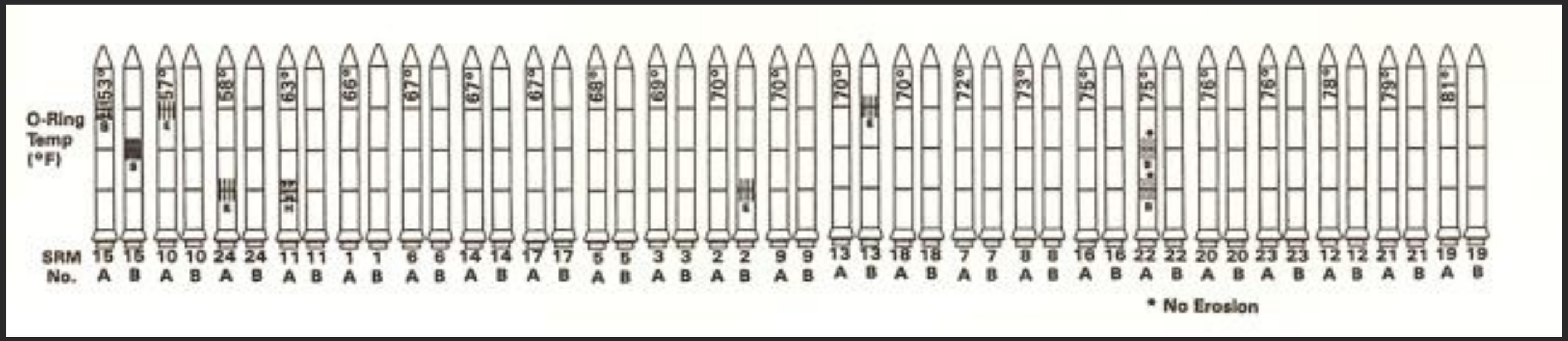
SRM-13A, 15, 16A, 18, 23A 24A
- NOZZLE BLOW-BY

HISTORY OF O-RING TEMPERATURES (DEGREES - F)

MOTOR	MBT	AMB	O-RING	WIND
DM-4	68	36	47	10 MPH
DM-2	76	45	52	10 MPH
QM-3	72.5	40	48	10 MPH
QM-4	76	48	51	10 MPH
SRM-15	52	64	53	10 MPH
SRM-22	77	78	75	10 MPH
SRM-25	55	26	29 27	10 MPH 25 MPH

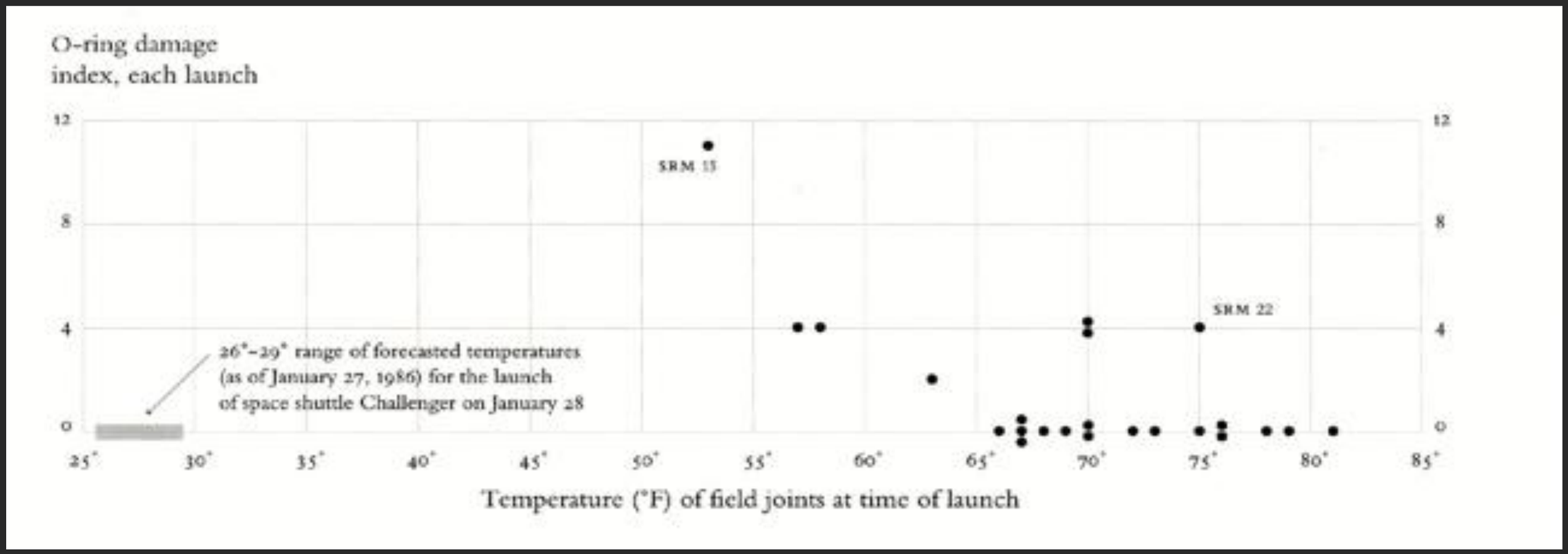
2 of 13 pages of material faxed to NASA by Morton Thiokol [from Tufte 1997]

Tell a story: Challenger



Visualizations drawn by Tufte show how low temperatures damage O-rings [Tufte 97

Tell a story: Challenger



Visualizations drawn by Tufte show how low temperatures damage O-rings [Tufte 97

Visualization Research

Challenge

More and more unseen data to understand

- Faster creation and collection
- Faster dissemination

But - the best displays are carefully created by domain experts who model task, weigh design alternatives, iterate – a time consuming process

Need better tools and algorithms for visually conveying information

Goals of visualization research

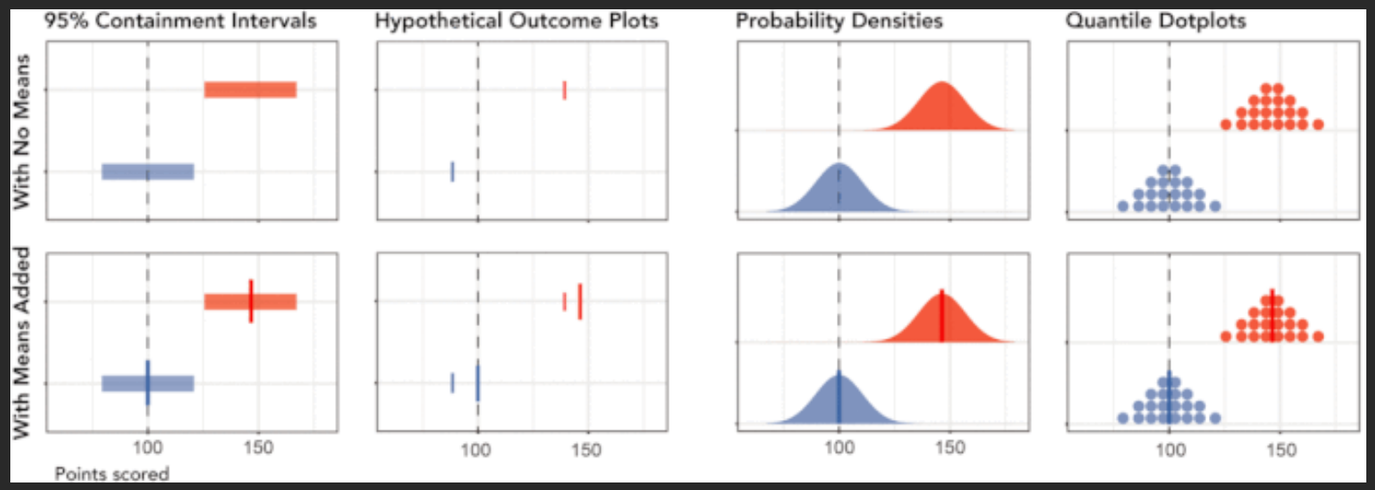
1. Understand how visualizations convey information to people

- What do people perceive/comprehend ?
- How do visualizations correspond with mental models of data?
- What are cognitively motivated design principles?

2. Develop techniques, systems for creating effective visualizations more easily

- Amplify perception and cognition
- Strengthen connection between visualization and mental models of data

Instructor: Alex Kale



Visualizing uncertainty

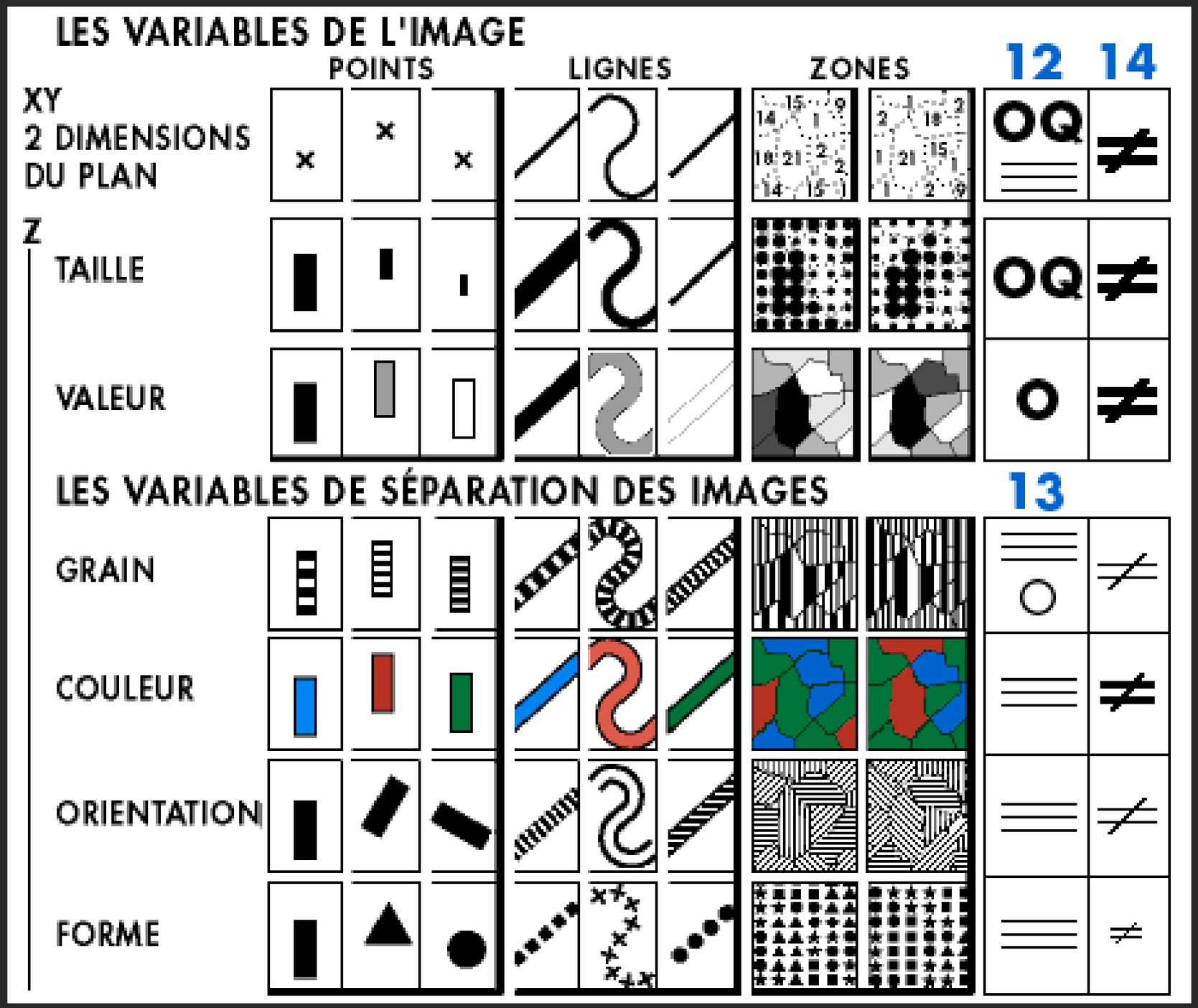
$$cs_A = \log\left(\frac{\mathcal{L}_A}{\mathcal{L}_B}\right) + \log\left(\frac{\theta_A}{\theta_B}\right)$$

Cognitive modeling

Statistical software

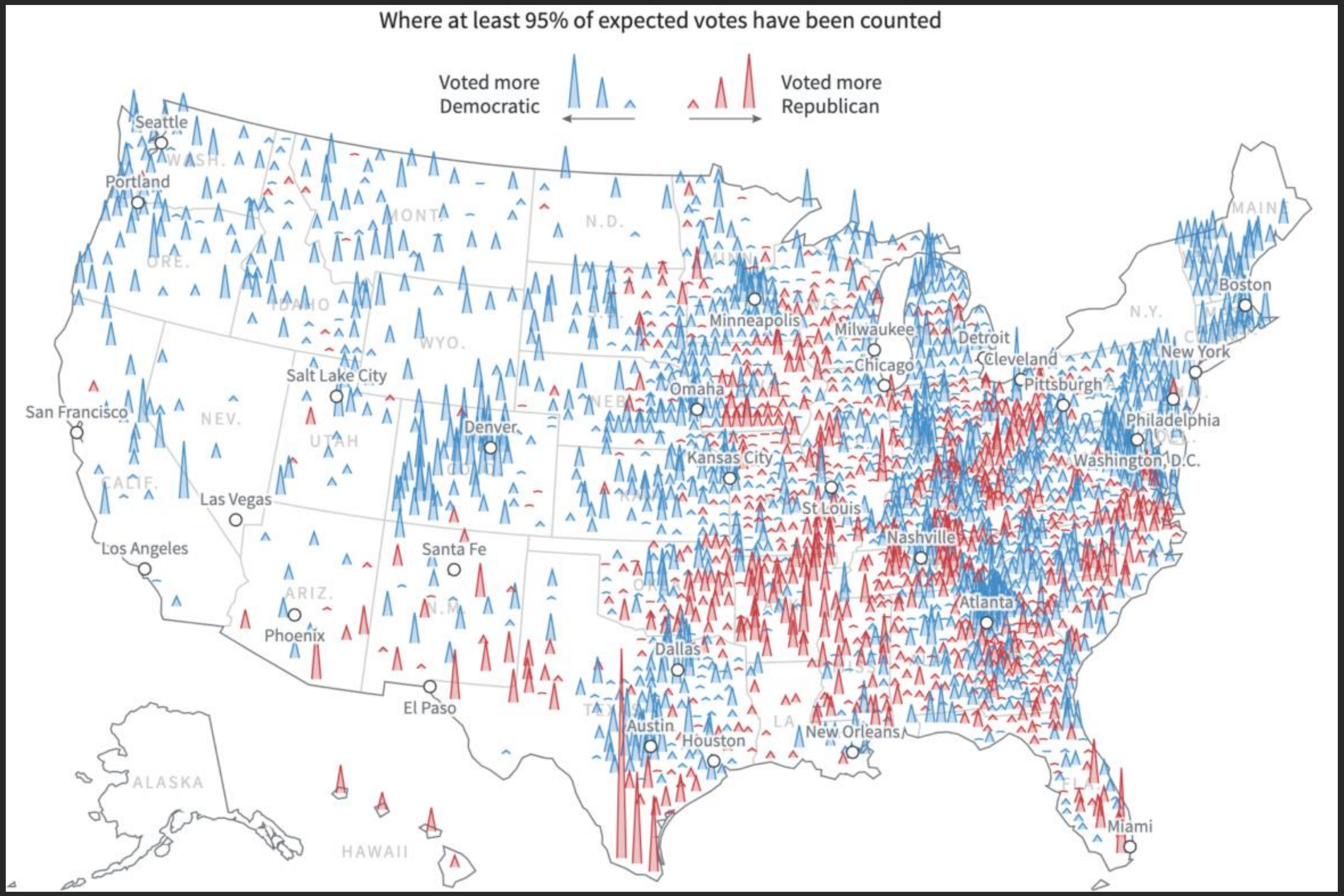
Topics

1. Vis fundamentals

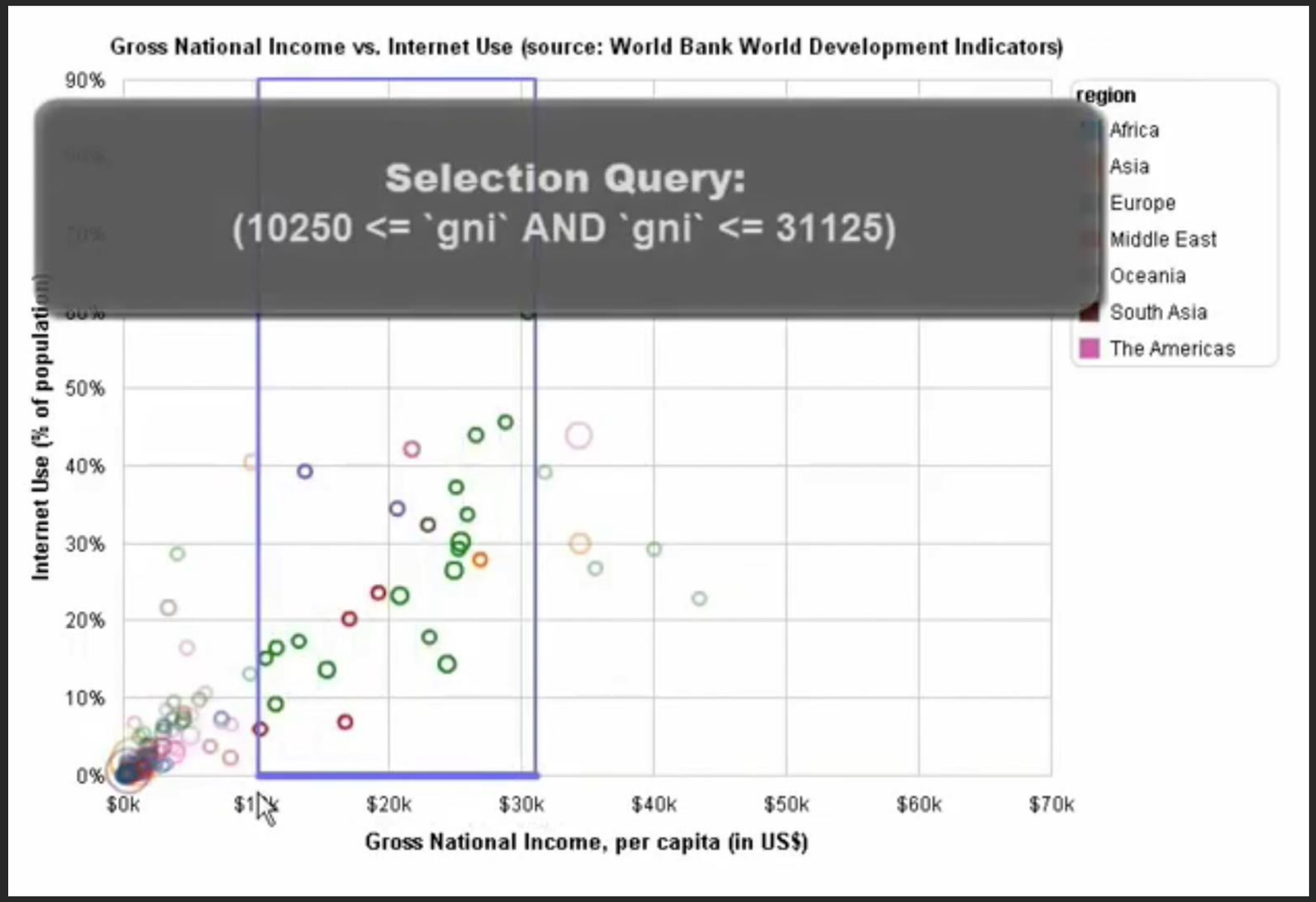


[Bertin, Graphics and Graphic Information Processing 1981]

2. Cartography, Svelte + D3

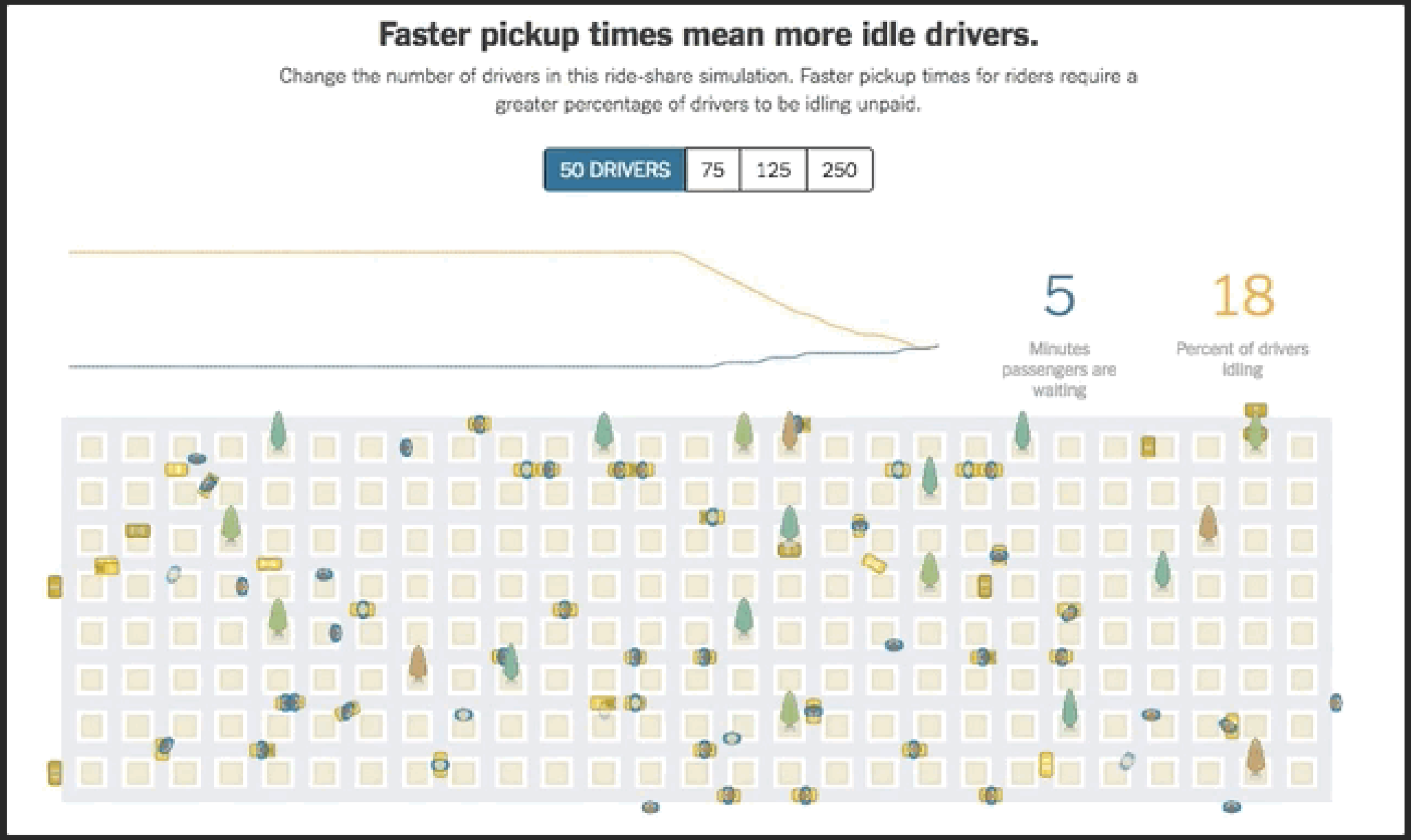


3. Interactivity and animation

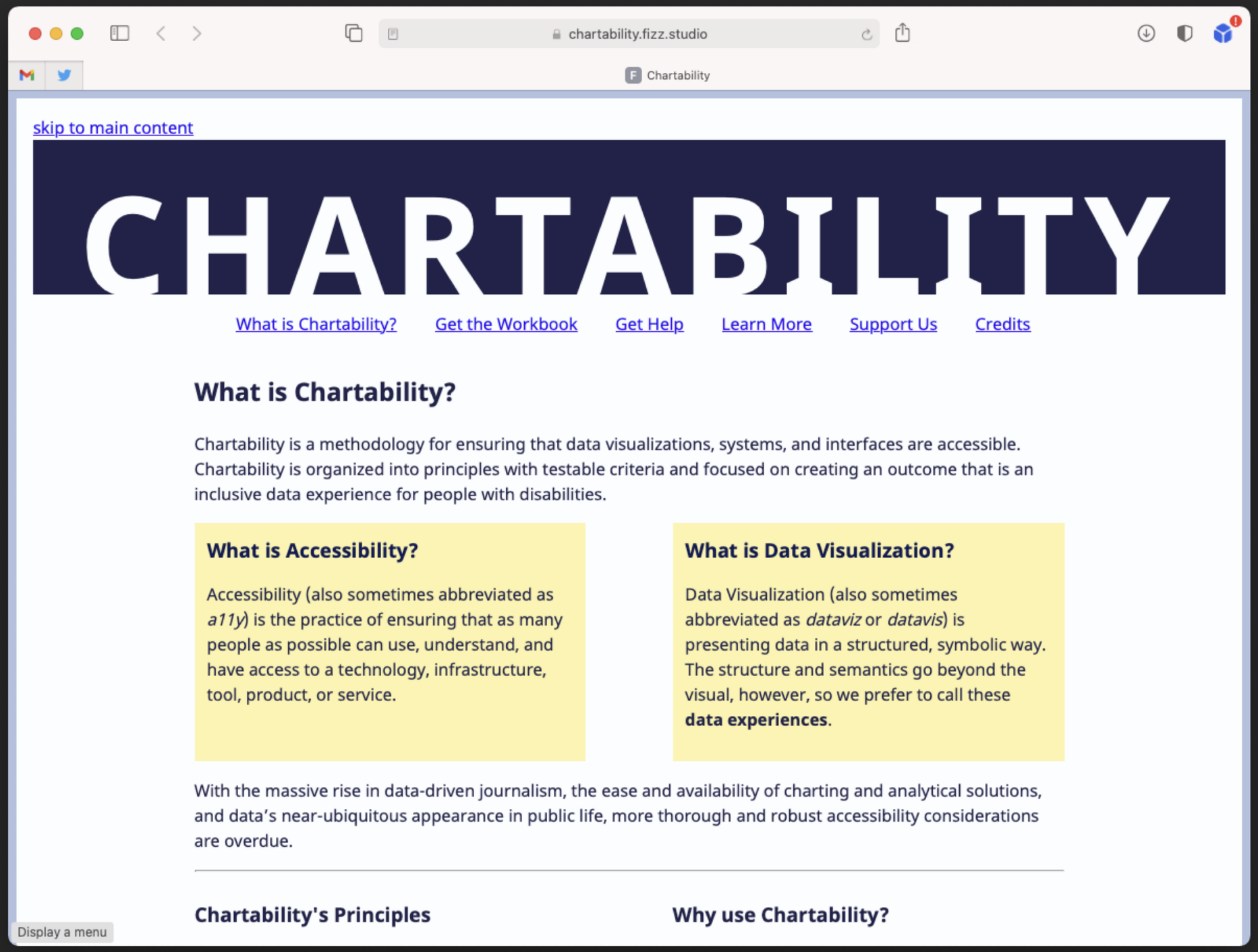


Generalized Selection [Heer et al. 2008]

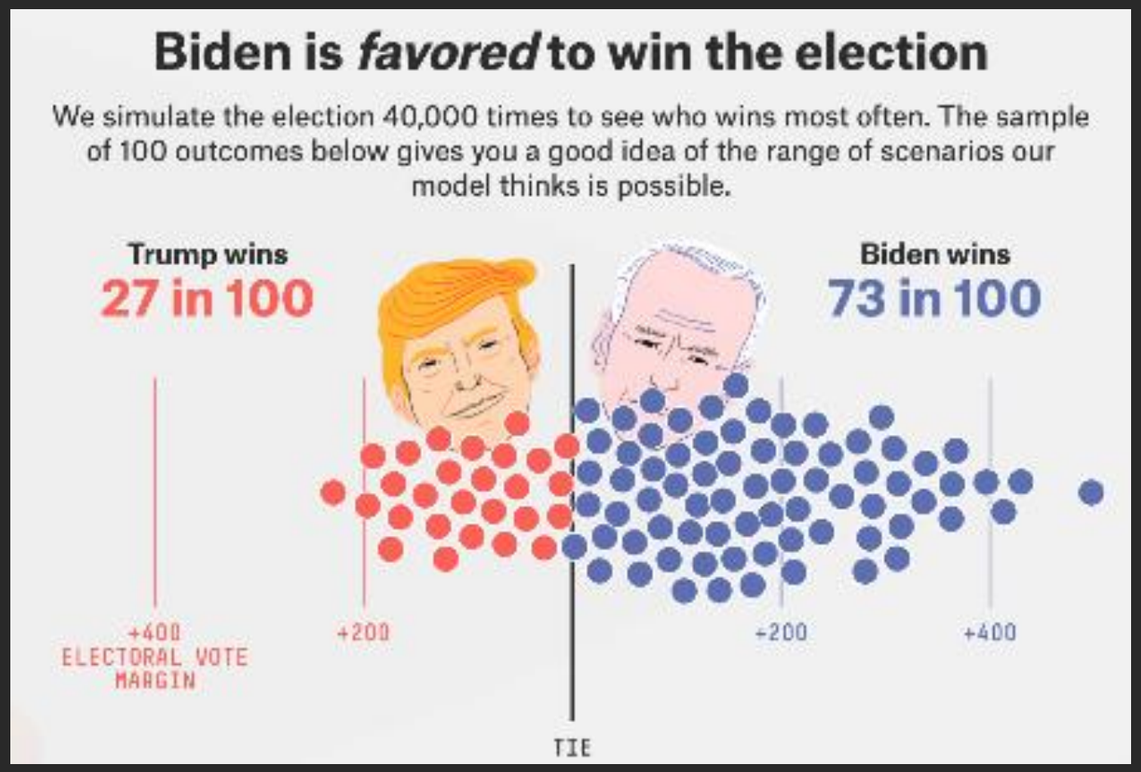
4. Data storytelling



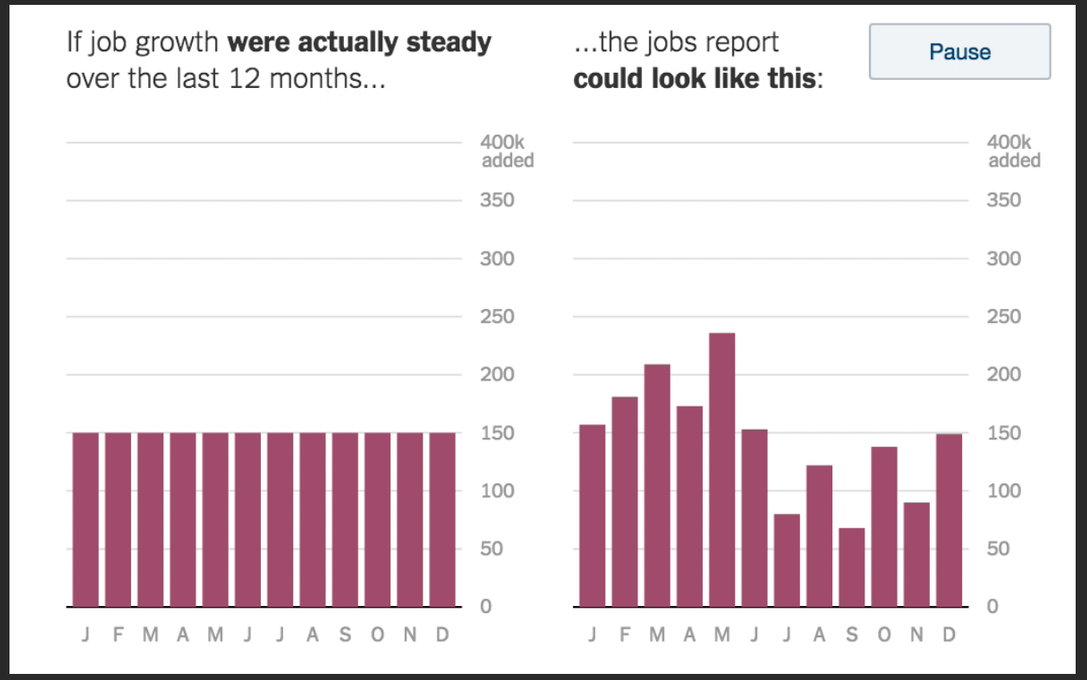
5. Accessibility



6. Visualizing uncertainty



Election forecast 2020 [FiveThirtyEight]



How Not To Be Misled By the Jobs Report [NYT]

Course Mechanics

Course Goals

1. Understand the mental model behind vis APIs and be able to use them.
2. Apply principles of perception, statistics, and human-centered design to data vis.
3. Make data interactive with JavaScript and JavaScript frameworks.
4. Use AI tools effectively to accelerate data exploration, analysis, design, and dev.
5. Use data interaction to advance and present original research.

Readings: All provided as PDFs

- No need to buy a textbook
- Readings are linked in the syllabus
- Check out resources.md on course website

Coursework

Assignments:

- Summative assessments of learning.
- Independent work, no collaboration.
- Opportunities for feedback and resubmission.

Project:

- Serves as a final.
- Groups of 3--4.
- Original research with a written proposal,, proof-of-concept, documented code, a presentation, and a written report.

Specifications grading

Satisfactory (S):

- Demonstrates mastery of material
- Very few minor mistakes or misjudgments
- No major errors

Needs improvement (N):

- Good faith effort, but lack of mastery
- More than a few minor mistakes, or a major error
- Partial credit, not available for exercises

Ungradable (U):

- Not submitted or incomplete
- Didn’t follow directions
- No credit

How to succeed in this course?

- Turn in all coursework on time!
- Start work early, budget time to debug/polish
- Spend time workshopping your writing
- Exercise statistical rigor and careful judgment
- Come to lecture, and pay attention
- Use resubmissions conservatively

Questions?

Assignment 1: EDA

Choose a dataset.

Perform an original exploratory analysis in a computational notebook.

Use a literate programming style to:

- Document your reasoning about analysis.
- Explain visualizations design choices.
- Interpret the charts you create.
- Present provisional take-aways.

Due April 3 at 11:59 PM

Project proposals

Possible high-level project ideas include *but are not limited to*:

1. Build a custom data collection interface using web technology
2. Create a public-facing data story, explorable explanation, or dashboard
3. Prototype a data-driven system or a design probe for user testing
4. Use visualization to advance scientific understanding of a complex dataset
5. Build an AI-assisted data analysis, processing, or sensemaking pipeline
6. Develop an interactive tool that combines visualization with AI capabilities

- Form groups of 3—4 (cannot change later)
- Identify original research problem and approach
- Draft motivating argument and proposal
- Break down the work required and delegate
- Must use vis, web tech, and/or AI to demonstrate learning

Due April 10 at 11:59 PM